

RAHUL MITRA

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EDUCATION

National University of Singapore

Singapore

Bachelor of Engineering, Industrial Systems Engineering

Aug 2023 – Jul 2027

- Awarded Certificate of Outstanding Performance in 3D Computer Vision (CS4277) — top student in a class of 24, NUS School of Computing.
- Second Major in Computer Science; Minor in Mathematics. Coursework: Software Engineering, Artificial Intelligence, 3D Computer Vision, Simulation, Statistics, Operations Research, Probability, Computer Systems.

Anderson Serangoon Junior College

Singapore

Singapore-Cambridge GCE A-Level

Feb 2019 – Dec 2020

EXPERIENCE

STMicroelectronics

Singapore

Operations Research (NUS System Design Project), Contract

Aug 2026 – Present

- Build an AI-driven put-away recommendation system for STMicroelectronics' Singapore warehouse, combining picking history, demand forecasts, and live capacity to replace experience-based slotting decisions.

Amazon

Singapore

Robotics Vision Engineer (BlendED AI+X)

Jul 2026 – Present

- Build vision systems for camera ISP enhancement, super-resolution, and image restoration under motion and low light.
- Engineer processing and evaluation stages that deliver perception-ready frame quality for downstream robotics vision.

Abbott Laboratories

Singapore

Operational AI Systems & Data Engineer (Contract)

Jun 2026 – Present

- Build an air-gapped process-intelligence platform for globally distributed spray-drying plants, spanning GPU OCR and layout parsing, recursive chunking, ONNX embeddings, a Chroma vector store, and LangChain over a locally hosted quantized Gemma model.
- Reverse-engineer four plants' APC steady-state optimizers from their Excel master files into a single HiGHS linear program with a tracking QP and PLC predictor, returning setpoints, binding constraints, and shadow prices at millisecond latency without a commercial solver license.

Abbott Laboratories

Singapore

Data Analytics and Project/Product Management Intern

Jan 2026 – Jun 2026

- Developed a SimPy discrete-event digital twin and OR-Tools CP-SAT optimizer for hybrid flow-shop scheduling, translating production objectives and constraints into decision-ready schedules.
- Engineered a non-destructive, 15-stage changeover-data pipeline to normalize operator- and machine-generated anomalies, validating it against five years of unseen raw data with zero execution failures.
- Productionized a team-built APC simulator for live manufacturing and engineer training, packaging it in Docker and deploying it to Azure App Service.

People's Association

Singapore

Software Engineer (Citizen Developer)

Sep 2025 – Present

- Built an end-to-end digital garden-plot balloting platform spanning admin and public frontends, backend, GIS mapping, and deployment, now production-ready for national rollout across Singapore.
- Integrated Singpass Login (OIDC) and Myinfo for passwordless authentication and consent-based retrieval of verified identity and registered address, automating resident eligibility checks.
- Led deployment planning, multi-repo coordination, and stakeholder management across product, engineering, and operations; awarded the S\$20,000 Sparks Community Innovation Fund.

YesWeHack

Remote

Independent Bug Bounty Researcher

May 2024 – Present

- Assessed GovTech GBBP12/13, the Land Transport Authority, and 13+ other programs for SSRF, CSRF, SQL/NoSQL injection, authentication bypass, and authorization weaknesses.
- Automated endpoint discovery, payload delivery, and evidence collection for responsible-disclosure reports using Burp Suite, Wireshark, Python, Bash, and custom fuzzing scripts.

Republic of Singapore Navy

Singapore

Base Support Assistant

Jan 2022 – Jan 2023

- Engineered an Excel VBA automation workflow that compressed a six-month food-waste analysis cycle into a sub-three-minute reporting process inside a security-constrained environment.
- Managed sensitive operational data under SAF confidentiality expectations while improving reliability, traceability, and analyst capacity.

PROJECTS AND COMPETITIONS

OnTheSpectrum — Codex & Blender MCP 3D Pipeline

May 2026

- Built a local-first Blender-to-Three.js asset pipeline using Codex and Blender MCP, generating GLB assets, metadata, previews, playable-world QA, and Remotion video workflows.

Hailo Edge AI Training and Deployment	Mar 2026
Trained and deployed OCR and object-detection models to Hailo edge accelerators, building the training and experimentation workflows around on-device inference and model acceleration.	
Arcane Security Tooling Family	Mar 2026
Built Rust-forward security and red-team tooling (Arcane, Arcane-PP, Arcane-GLM, Arcane-OCR) for automation, OCR, and adversarial workflows.	
Waaah Comics — 24-Hour Gesture-to-Comic Build	Jan 2026
Built a gesture-to-comic pipeline in a 24-hour sprint, turning MediaPipe 3D hand landmarks into generated comic strips with Gemini 2.0 Flash, Veo 3, and Konva.js canvas tooling.	
Volt Pulse SG — SMU Hack For Cities 2026 (Top 8 Finalist of 50+ teams)	Jan 2026
Built an agentic AI system for tracking HDB household energy costs using SEALION embeddings, Supabase vector search, RRF routing, and scheduler-agent planning.	
Maritime Deficiency Severity Forecasting — Maritime Hackathon 2025 (Team Lead)	2025
Led model training for a maritime-inspection NLP system, fine-tuning a 109M-parameter BERT with DNN layers, gradient clipping, and hyperparameter optimization on the ASPIRE 2A supercomputer.	
AgeWellLah.AI — NUS HealthHack 2025	2025
Built an elderly-care triage chatbot using OpenAI GPT-4, InterSystems IRIS Vector Search, and Flask, with OAuth2-secured patient profiles and PDPA-aware design.	
SmartExam — GovTech x NTUPC Product Hackathon	2025
Built a multi-agent autonomous exam-generation system with a RAG pipeline and Next.js frontend, automating source ingestion, question generation, and education workflows.	
EthosLens — LifeHack 2025	2025
Built an AI research engine scoring brand ethics and sustainability using FastAPI, Python, and LangChain with automated evidence retrieval.	
DSTA BRINHACK 2025 CTF (top quartile of 400+ teams)	2025
Solved web, crypto, reverse-engineering, and systems-security challenges under time pressure in a first CTF, finishing in the top quartile among 400+ teams.	
Additional Projects	2023 – 2026
- NVIDIA Disaster Risk Monitoring with Satellite Imagery — geospatial AI with U-Net semantic segmentation on GPU/ASPIRE 2A workflows (2023). - Fine-tuning LLMs for Cybersecurity — coursework exploring Mistral/LLaMA fine-tuning with AutoTrain and AutoGen (2024).	

LEADERSHIP AND ACTIVITIES

NTUC Health Active Ageing Centre	Singapore
<i>Elderly Care Volunteer</i>	Sep 2025 – Present
Support weekly senior-engagement activities, deepening empathy for healthcare, accessibility, and human-centered community product design.	
STEM Inc. (ASRJC)	Singapore
<i>Astronomy Head & Robotics Member</i>	Mar 2019 – Aug 2020
Led astronomy outreach, competed in the 7th Singapore Astronomical Olympiad, engineered an autonomous robot for the National Robotics Challenge, and received the ASRJC Outstanding Contribution Award.	

SKILLS AND CERTIFICATIONS

Languages: Python, TypeScript, JavaScript, Java, Rust, Bash, C++, C#, Excel VBA, SQL

AI/ML: BERT, DNN, transformers, PyTorch, RAG, AI agents, GPT-4, Gemini, SEALION, LangChain, ONNX, Chroma, local LLM serving (llama.cpp GGUF), vector search, NLP, deep RL (PPO, A2C, DDPG, DQN), model evaluation; vision: camera ISP, super-resolution, image restoration, 3D CV (projective & epipolar geometry, absolute pose, SfM, bundle adjustment, multi-view stereo); HPC: ASPIRE 2A, CUDA, distributed training

Operations Research: CP-SAT, HiGHS linear and quadratic programming, SciPy, hybrid flow-shop scheduling, SimPy, digital-twin simulation, signal processing (Kalman, Welch PSD, change-point, matrix profile), robust optimization, graph optimization, network flow, Dijkstra/SSSP/APSP, decision analytics, statistics, mathematical modeling

Cloud/DevOps: AWS (Fargate, Route 53), Terraform, Azure, Docker, Redis, Kafka, CI/CD, Bitbucket CI/CD, GitHub Actions, Supabase, API design, Codex

Security: Burp Suite, Wireshark, SSRF, CSRF, SQL/NoSQL injection, authentication bypass, OAuth2/JWT, OWASP Top 10, secure design, fuzzing, reverse engineering, custom exploit scripting, cloud security (AWS IMDSv2, metadata APIs)

Web/Full-Stack: React, Next.js, Three.js, Matter.js, FastAPI, Flask, aiohttp, asyncio, GIS mapping, Singpass Login & Myinfo (OIDC), Konva.js

Architecture/Product: Solution architecture, manufacturing systems, stakeholder discovery, rapid prototyping, accessibility, product discovery, user testing, dashboards

Hardware/Embedded: Edge AI / Hailo model deployment (OCR, object detection), autonomous robotics (sensor integration)

Certifications: NVIDIA Disaster Risk Monitoring with Satellite Imagery; Deep Reinforcement Learning Hands-On (Packt); Docker for the Absolute Beginner; Reverse Engineering Windows Executables; Excel 2019 Advanced; Deep Learning CNN and RNN with Python (Packt); Learning C# by Developing Games with Unity (Packt)